Electronic supplementary material

Payoff scale reshapes how language models play
the prisoner's dilemma

Contents

S1 Scope of the corpus

The analysis in the main text uses one corpus of language-model transcripts, `Dataset/data_fairgame_frontier_llm`, and one synthetic corpus, `Dataset/noise_dataset`, which trains the LSTM branch of the strategy read-out and contains no language-model output at all. No other dataset in the repository enters any number reported in the paper.

The transcript corpus is a complete crossing of ten payoff scales, six models, five prompt languages, four persona pairings and ten replicates, at ten rounds per game and two agents per dyad. Table S1 gives its composition. Every one of the 300 model-by-language-by-scale cells holds exactly 40 dyads, hence 80 agent-games and twenty decisions per dyad, or 200 dyads and 400 agent-games per cell of the coarser model-by-scale grid, and the ingest raises an error rather than writing its tables if that is not what is on disk. The corpus is therefore 12,000 dyads, 24,000 agent-games and 240,000 decisions.

The main text reports five of the six models and this supplement reports all six. The sixth, Gemini 3.1 Flash-Lite, is the subject of §S8, which states why it sits here rather than in the main text and gives the all-six version of every pooled quantity the main text reports on five. The five carry 250 of the 300 cells, 10,000 dyads, 20,000 agent-games and 200,000 decisions.

The completeness guard summarized in the main text has five clauses, and the build refuses to write its tables if any of them fails: a payoff scale is missing; a cell departs from 40 dyads; the cell count is not the product of the factor levels; an agent-game is not ten rounds long; or a model directory was not registered in advance, which is what stops an unrecognised model being skipped in silence. It also recovers each game's payoff scale from the payoffs recorded in it, as the median ratio of realised to base payoff, and refuses if that disagrees with the scale the cell was run at by more than a relative 10^{-6}. That last check catches a mislabelled directory.

The action mapping was checked against the data as well as against the prompt. Reproducing a published cooperation curve from these tables gives a mean absolute error of 0.002 under the mapping used here, in which OptionB is cooperation, and 0.171 under the inverted one.

S2 Protocol and collection detail

This section carries the collection detail that Materials and methods summarizes: how the models were served, how the prompt files were checked, what the agent was and was not told, and the seeds, parsing and resampling the analysis rests on.

All six models were served through the Kaggle Benchmarks model proxy on an OpenAI-compatible chat-completions endpoint, software development kit version 0.6.1. The endpoint identifiers were `claude-haiku-4-5-20251001` for Claude Haiku 4.5, `gpt-5.4-nano-2026-03-17` for GPT-5.4 Nano, `gemini-3.5-flash-lite` for Gemini 3.5 Flash-Lite, `qwen3-235b-a22b-instruct-2507` for Qwen3 235B-A22B, `grok-4.20-0309-non-reasoning` for Grok 4.20 and `gemini-3.1-flash-lite-preview` for Gemini 3.1 Flash-Lite; the two Gemini identifiers name endpoints that are not pinned to a release date, which is why the second of them is reported here rather than in the main text (§S8).

The prompt templates are compared byte for byte against the FAIRGAME resource files before any collection run because one save in the wrong encoding corrupts the prompt that reaches the model rather than a comment beside it, and the corruption is invisible in the recorded results. Artefacts of the resource files are reproduced rather than repaired. Two of them are visible in the prompts used here: the Arabic and Chinese versions state the four cells as penalties while phrasing the goal sentence as maximizing a reward, and the Vietnamese version doubles an article in the persona line.

The opponent's persona is never disclosed: the protocol gates that disclosure behind a probability, which is set to zero here, and the block naming the opponent is deleted from every prompt.

S2.1 Seeds

The seed sent with each request is a deterministic function of the base seed 12345 and an index over prompt language, persona pairing and replicate, the round number, and which of the two agents is replying. The index excludes λ, so the ten scales share one stream of seeds. A seed on a hosted endpoint is honoured on a best-effort basis rather than guaranteed, and sampling runs at temperature 1.0, so the scheme is variance reduction and not determinism.

S2.2 Parsing and fallback

Decisions are elicited as free text, with no response schema and no function calling, under a ceiling of 128 output tokens and no reasoning-effort parameter. Replies are parsed by the protocol's own matcher, which normalizes to lower-case alphanumerics and searches in turn for the two option names, for the internal strategy keys, and finally for a bare letter or digit. A reply it cannot resolve is regenerated with a different seed, up to two further attempts, so one decision costs at most three generations. The achieved parse-failure and fallback rates are held in the collection logs rather than in the released per-game tables; each run asserts that its fallback rate is at most 0.02 and fails otherwise. Because those rates are not in the released tables we cannot report how they vary with the payoff scale, and a rate that rose with the printed magnitudes would bias the smallest ranges downward.

S2.3 Resampling and permutation

One seed, 20260905, drives every resampling reported in the paper, so intervals in different rows of a table are drawn against common numbers.

The permutation test shuffles the scale label within strata of prompt language by persona pairing. The unit shuffled is the dyad rather than the agent-game, because the payoff scale was assigned to a whole dyad: under the null both of its agent-games carry the same label, exactly as they do in the data. Shuffling agent-games independently would break that pairing and draw from a null narrower than the true one, which would make the p-value anti-conservative. It is the same dependence that makes every interval in the paper resample whole dyads and every regression cluster on them. Because the two sides of a mixed pairing are one dyad seen twice, the strata are the five prompt languages crossed with the three dyad types, cooperative against cooperative, selfish against selfish and the mixed pair, giving 15 strata inside each model: five of 100 dyads, five of 100 and five of 200 respectively.

S3 Cooperation, by scale, model and language

Table S2 gives the cooperation rate behind figure 1b of the main text, with bootstrap intervals. Table S3 breaks the same quantity down by prompt language and is the table behind figure 2.

The final column of table S3 is the quantity we draw attention to in the main text: the range of cooperation across the ten payoff scales, computed within one model and one language. Over all six models it runs from 0.080 (Gemini 3.1 Flash-Lite in Arabic) to 0.855 (Grok 4.20 in English); over the five the main text reports, the smallest is 0.083 (Qwen3 235B-A22B in French) and the largest the same 0.855. The sensitivity of a model to a transformation that changes nothing about the game is therefore not a single number even for a fixed model; it depends on the language the game was posed in.

S4 Variance decomposition

Table S4 is the full analysis-of-variance table summarized in the main text, on the 250 cell means of the five models reported there. Table S5 is the same decomposition on all 300 cell means, the sixth model included. Both are computed on cell means rather than on individual agent-games, because the design is exactly balanced at the cell level, which makes the type-II decomposition order-independent.

The reading that matters for benchmark practice is the contrast between rows, and it is the same in both tables. On the five models the payoff scale carries 5.8% of the variance across cells as a main effect and is significant ($p < 0.001$), while its interaction with the model carries 17.9%, roughly three times as much, and its interaction with the prompt language 3.6%. On all six the corresponding figures are 4.3%, 19.1% and 3.1%, every term again significant. Most of what the payoff scale does is therefore specific to the model rather than shared across models, and a design that estimates only a main effect, pooling over models and languages, reports a number that describes no model in the pool.

The main effect is significant in these panels because one model, Grok 4.20, departs from the invariance far enough to survive being averaged with the others. The same decomposition run on the five models that remain when it is removed returns $F_{9,144} = 1.10$, $p = 0.36$ and 0.7% of the variance for the payoff scale as a main effect, which is the null, while each of those five models is separately and significantly sensitive to the scale, by both the permutation test and the clustered Wald test (table 1 of the main text for four of them, §S8 for Gemini 3.1 Flash-Lite). Whether a pooled design finds an average effect here is a fact about which models are in the panel.

S5 Persona

Table S6 gives the persona effect at every scale and model, which is the table behind figure 3a. The two leftmost columns are the scales at which every payoff printed is at most 1, since with base payoffs $0, 2, 6, 10$ the largest number shown is 10λ.

Two features are worth stating explicitly because they are easy to misread from the figure. First, the sign reversal is a reversal of the *persona* effect, not of the cooperation rate: below the boundary the two reversing models cooperate more when told they are selfish than when told they are cooperative. Second, two models are inverted at every scale rather than only below the boundary. Qwen3 235B-A22B is inverted by a wide margin, cooperating 0.072 under the cooperative persona against 0.798 under the selfish one, and Grok 4.20 by a narrower one. We report it because excluding it would flatter the analysis, but the paper draws no conclusion from the inversion itself beyond noting that its magnitude, like everything else measured here, depends on the payoff scale.

S6 Strategy read-out

S6.1 What the two branches do

Each agent-game is encoded as a sequence of two-letter tokens, one per round, pairing the previous round's joint outcome as the focal player saw it with the action taken in the current round; with padding the vocabulary is eleven tokens. The alphabet is shared between the synthetic training corpus and the language-model transcripts, which is what makes a classifier trained on the former applicable to the latter without any adaptation.

The rule base asks an exact question: is this trajectory precisely what AllC, AllD, Tit-for-Tat or Win-Stay-Lose-Shift would have played, given the history that actually occurred? Because all four rules are memory-one, the answer is a per-token lookup and requires no learning. It returns a set rather than a single label, because over ten rounds several canonical rules routinely prescribe the same moves. A player cooperating in all ten rounds against a cooperator is exactly consistent with three of the four rules at once, AllC, Tit-for-Tat and Win-Stay-Lose-Shift, and no method, learned or otherwise, can separate them on that evidence. Reporting the set is the correct answer, not a failure.

The LSTM ranks within whatever set the rule base returns, and decides alone only where the set is empty. It therefore cannot overturn a deduction; it can only break a tie the evidence genuinely leaves open, or supply an attribution where no canonical rule applies.

S6.2 Provenance, and which claims depend on the classifier

Over the whole corpus, 23.0% of agent-games are matched exactly by one canonical rule, 17.5% by several at once, and 59.6% by none. Those three provenances are reported separately throughout, and the claims in the main text are ordered by how much they depend on the learned component:

- The share of agent-games matched exactly by one rule (table S7) and the number of rounds violating the nearest canonical rule are computed by exact matching. Neither depends on the LSTM in any way. These carry the main text's strategy claim.

- The compositional claim, that the mix of labels moves with the payoff scale and moves differently in different models (table S8), uses the labels, and therefore leans on the LSTM for the 77.0% of agent-games that no single rule matches exactly: it decides alone for the 59.6% that match no rule at all, and ranks within the matcher's own set for the remaining 17.5%. It is the weaker of the two and is presented as such.

Of the agent-games carrying the AllC label, 23.6% were deduced from an exact rule match and a further 36.9% came from a rule set the LSTM ranked within; for AllD the deduced share is 35.6%.

Table S9 shows that this dependence is very unevenly spread across the four labels, which bounds how the compositional result may be read. AllC and AllD carry deduced shares of 23.6% and 35.6%, so a movement in those two is anchored in part by exact matching. Tit-for-Tat and Win-Stay-Lose-Shift are 94.6% and 97.1% unmatched, meaning the label is an attribution the LSTM makes about which canonical rule the play most resembles, on trajectories that follow none of them exactly. A rise in the Tit-for-Tat share is therefore evidence that play is drifting towards the reciprocal region of the space, and is not evidence that any model is implementing Tit-for-Tat. The main text states this where the result is reported.

S6.3 Validation

Table S10 reports the accuracy of the LSTM branch. Training uses the noise-free and 5% execution-noise variants of the synthetic corpus together, balanced at 40,320 sequences per strategy per level on a random 80/20 split, which forces the network to identify a rule from imperfect behaviour rather than from a clean signature; the 10% and 20% variants are held out entirely and measure how the read-out degrades as play departs further from any rule.

The network is a single-layer LSTM with 24-dimensional embeddings, 96 hidden units and dropout 0.15, read at its last valid step. The optimiser is AdamW at learning rate 3×10^{-3} and weight decay 10^{-4}, with a cosine schedule over 12 epochs, batches of 512, gradient-norm clipping at 1.0 and seed 1234.

Accuracy falls from 98.5% in distribution to 93.6% and 81.1% at those two unseen noise levels, which is the honest bound on how far the label distributions in table S8 should be pushed.

S7 Why the model curves cancel under pooling

Table S11 gives the pairwise correlations between the five main-text models' cooperation-against-scale curves. Six of the ten pairs are negative, the mean is -0.10, and the extremes are -0.72 and $+0.85$. Averaging curves that disagree in shape this strongly reduces the range from a mean of 0.249 within a model to 0.154 for the average, a factor of 1.62. The cancellation is arithmetic rather than statistical, and no increase in sample size affects it.

Table S12 repeats the calculation on all six models. Nine of the fifteen pairs are negative, the mean correlation is -0.08, and the extremes are unchanged at -0.72 and $+0.85$. The pooled range falls to 0.125 against a mean within-model range of 0.219, an attenuation of 1.76: adding the sixth model makes the cancellation slightly worse, not better.

Cancellation of this kind is partial rather than total here, and the main text says so. A significant main effect of the payoff scale survives the averaging in both panels, and both facts belong in the same sentence: the average is non-zero, and at 0.154 against a mean within-model range of 0.249 it matches the movement of no model in the average, understating the three that move most and overstating the two that move least.

S8 The sixth model, Gemini 3.1 Flash-Lite

The corpus holds six models. The main text reports five of them and this supplement reports all six, and every table above already carries the sixth alongside the rest. This section states why it sits here and what including it changes.

S8.1 Why it is reported here

The reason is a property of the endpoint and not of the results. Gemini 3.1 Flash-Lite was served under the identifier `gemini-3.1-flash-lite-preview`, which names a preview endpoint that the provider does not pin to a fixed version. What was served under that name while the corpus was collected cannot be recovered afterwards, and a reader cannot re-run the same object. The other five identifiers name endpoints we can re-address, so they are stable objects of study in a way this one is not, and the main text is restricted to them.

We state the consequence plainly, because the reader is otherwise asked to take the split on trust. On one measure this model is the only counter-example in the corpus. The share of its agent-games matched exactly by one canonical rule *rises* with the payoff scale, from 17.8% below the sub-unit boundary to 26.3% above it ($\beta = +0.125$ per decade of λ, $p < 0.001$), where the other five either fall or show no trend. Its play therefore becomes more rule-like as the payoffs grow, and the main text says so where that result is reported rather than leaving it to be found here. On the continuous version of the same measure, the number of rounds violating the nearest canonical rule, it has no significant trend at all ($\beta = -0.019$, $p = 0.26$), so the counter-example is confined to the label-based measure.

S8.2 What including it changes

Nothing that this paper concludes. The comparison is set out term by term above and can be read directly off the tables:

- **Departure from the invariance.** Its range across the ten scales is 0.072 (95% CI 0.060, 0.133), the smallest in the corpus, with $p = 0.0005$ from the same permutation test and

$\chi^2_9 = 83.1$, $p < 0.001$, from the same clustered Wald test on the nine scale contrasts; its cooperation at each scale is in table S2. It departs from the invariance like every other model, and departs by less than any of them.

- **Pooling.** With all six models the pooled range is 0.125 against a mean within-model range of 0.219, an attenuation of 1.76, and nine of the fifteen pairwise shape correlations are negative with a mean of −0.08 (§S7). The five-model figures are 0.154, 0.249, 1.62 and six of ten negative. The conclusion is the same and marginally stronger with the sixth model in.

- **Variance decomposition.** On all 300 cell means the model carries 44.8% of the variance, the payoff scale 4.3% as a main effect, the model-by-scale interaction 19.1%, the model-by-language interaction 18.0% and the language-by-scale interaction 3.1%, every term significant at $p < 0.05$ (table S5). The ordering is that of the five-model table.

- **Persona.** Its persona effect shifts by 0.103 (95% CI 0.075, 0.131) across the sub-unit boundary, which excludes zero as in every other model, and it does not change sign: it is positive in both regimes, +0.577 below the boundary and +0.680 above it. The count of sign reversals in the main text, two models of five, becomes two of six.

- **Opening move.** Its range of opening-move cooperation across the ten scales is 3.68 times its range over rounds 2 to 10, the largest ratio in the corpus. The main text's four models of five with a ratio above one becomes five of six.

The one count that changes qualitatively is the strategy trend, and it changes in the direction that weakens rather than strengthens the paper's claim: four models of five moving one way with no counter-example becomes four of six moving one way, one showing no trend and one moving the other. That is stated in the main text where the result is reported and again in its limitations.

S9 Reproduction

The pipeline runs in order and each stage writes its outputs under `Analysis/scaling/`:

- `s00_build.py` parses the transcripts into `rounds.parquet` and `games.parquet`, and refuses to write either if the grid is incomplete.

- `s01_train_lstm.py` trains the LSTM branch on the synthetic corpus and writes `models/strategy_lstm.pt`.

- `s02_readout.py` applies the hybrid read-out to all 24,000 agent-games and writes `readout.parquet`.

- `s03_stats.py` and `s04_strategy_stats.py` compute every statistic in the paper into `tables/`.

- `s05_figures.py` draws figures 1 to 6.

- `s06_tables.py` and `s07_supplementary.py` generate the LaTeX tables in the main text and in this supplement.

- `s08_verify_paper.py` re-derives every number quoted in the prose of the main text from the tidy tables and fails if any disagrees.

Random seeds are fixed: 1234 for classifier training, 20260905 for the bootstrap and the permutation test.

Analyses were run in Python 3.12.3 with `numpy` 1.26.4, `pandas` 2.2.2 and `statsmodels` 0.14.6 for the clustered logistic regressions and the type-II analysis of variance, `matplotlib` 3.9.2 for the figures, and `pytorch` 2.11.0 for the strategy classifier.

S10 Supplementary tables

Table S1: The corpus, by model. Every model was run on the same ten payoff scales and five prompt languages. "Cell n" is the number of dyads in each model × language × scale cell; the design is balanced with none missing, and the build refuses to write its tables otherwise. The first five models are the ones the main text reports; Gemini 3.1 Flash-Lite is reported here, for the reason given in §S8.

Model	Scales	Lang.	Dyads	Agent-games	Decisions	Cell n	Coop.
Claude Haiku 4.5	10	5	2,000	4,000	40,000	40	0.415
GPT-5.4 Nano	10	5	2,000	4,000	40,000	40	0.606
Gemini 3.5 Flash-Lite	10	5	2,000	4,000	40,000	40	0.751
Qwen3 235B-A22B	10	5	2,000	4,000	40,000	40	0.435
Grok 4.20	10	5	2,000	4,000	40,000	40	0.713
Gemini 3.1 Flash-Lite	10	5	2,000	4,000	40,000	40	0.617
All	10	5	12,000	24,000	240,000	40	0.590

Table S2: Cooperation by model and payoff scale. Each entry aggregates 200 dyads. The bracketed line under each model is a 95% confidence interval from a bootstrap that resamples whole dyads.

Model	λ=0.01	λ=0.1	λ=0.25	λ=0.5	λ=1	λ=2	λ=5	λ=10	λ=100	λ=1000
Claude Haiku 4.5	0.407	0.379	0.371	0.427	0.452	0.444	0.424	0.435	0.407	0.401
	[0.37,0.45]	[0.34,0.42]	[0.34,0.40]	[0.39,0.46]	[0.42,0.49]	[0.41,0.48]	[0.39,0.46]	[0.40,0.47]	[0.38,0.43]	[0.37,0.43]
GPT-5.4 Nano	0.438	0.540	0.644	0.598	0.576	0.637	0.680	0.633	0.663	0.653
	[0.40,0.48]	[0.50,0.58]	[0.61,0.68]	[0.56,0.63]	[0.54,0.61]	[0.60,0.67]	[0.65,0.71]	[0.60,0.67]	[0.64,0.69]	[0.62,0.68]
Gemini 3.5 Flash-Lite	0.792	0.694	0.872	0.800	0.774	0.710	0.697	0.754	0.734	0.682
	[0.74,0.84]	[0.64,0.74]	[0.85,0.89]	[0.77,0.83]	[0.75,0.80]	[0.67,0.74]	[0.66,0.73]	[0.72,0.78]	[0.70,0.77]	[0.64,0.72]
Qwen3 235B-A22B	0.476	0.494	0.455	0.432	0.416	0.427	0.419	0.422	0.399	0.411
	[0.43,0.52]	[0.44,0.54]	[0.41,0.50]	[0.39,0.47]	[0.38,0.45]	[0.39,0.46]	[0.38,0.45]	[0.39,0.45]	[0.37,0.43]	[0.38,0.44]
Grok 4.20	0.274	0.440	0.815	0.630	0.809	0.910	0.855	0.877	0.654	0.865
	[0.23,0.32]	[0.39,0.49]	[0.77,0.85]	[0.59,0.67]	[0.77,0.85]	[0.88,0.94]	[0.82,0.89]	[0.84,0.91]	[0.61,0.70]	[0.83,0.90]
Gemini 3.1 Flash-Lite	0.641	0.648	0.621	0.640	0.635	0.614	0.614	0.598	0.584	0.577
	[0.61,0.67]	[0.62,0.68]	[0.59,0.65]	[0.61,0.67]	[0.61,0.67]	[0.58,0.65]	[0.58,0.65]	[0.56,0.63]	[0.54,0.62]	[0.54,0.61]

Table S3: Cooperation by model, prompt language and payoff scale. 40 dyads per entry. The final column is the range across the ten scales for that model and language; over all six models it runs from 0.080, for Gemini 3.1 Flash-Lite in Arabic, to 0.855, for Grok 4.20 in English, so the scale sensitivity of a model is a property of the model and the language jointly.

Model	Language	0.01	0.1	0.25	0.5	1	2	5	10	100	1000	Range
Claude Haiku 4.5	English	0.35	0.40	0.32	0.42	0.44	0.45	0.41	0.42	0.45	0.46	**0.136**
	French	0.65	0.47	0.50	0.33	0.37	0.43	0.41	0.40	0.38	0.40	**0.314**
	Vietnamese	0.17	0.17	0.28	0.29	0.26	0.20	0.30	0.22	0.23	0.23	**0.130**
	Chinese	0.53	0.52	0.45	0.61	0.69	0.57	0.56	0.66	0.58	0.56	**0.245**
	Arabic	0.35	0.33	0.31	0.48	0.51	0.56	0.43	0.47	0.40	0.36	**0.249**
GPT-5.4 Nano	English	0.45	0.64	0.85	0.88	0.86	0.90	0.88	0.87	0.89	0.76	**0.449**
	French	0.75	0.78	0.78	0.57	0.55	0.60	0.62	0.67	0.55	0.74	**0.226**
	Vietnamese	0.49	0.46	0.58	0.47	0.50	0.62	0.75	0.54	0.65	0.63	**0.289**
	Chinese	0.24	0.45	0.65	0.60	0.49	0.57	0.61	0.58	0.62	0.65	**0.414**
	Arabic	0.25	0.37	0.37	0.48	0.47	0.49	0.54	0.51	0.61	0.49	**0.360**
Gemini 3.5 Flash-Lite	English	0.94	0.66	0.92	0.84	0.74	0.59	0.53	0.65	0.64	0.60	**0.412**
	French	0.91	0.63	0.86	0.72	0.78	0.63	0.75	0.78	0.59	0.61	**0.317**
	Vietnamese	0.72	0.63	0.93	0.86	0.81	0.76	0.62	0.64	0.89	0.73	**0.314**
	Chinese	0.69	0.80	0.76	0.64	0.70	0.73	0.67	0.76	0.66	0.73	**0.162**
	Arabic	0.70	0.75	0.89	0.94	0.83	0.84	0.92	0.93	0.89	0.74	**0.242**
Qwen3 235B-A22B	English	0.47	0.51	0.53	0.44	0.42	0.49	0.43	0.50	0.37	0.46	**0.155**
	French	0.41	0.40	0.35	0.39	0.37	0.36	0.35	0.35	0.34	0.33	**0.083**
	Vietnamese	0.52	0.55	0.48	0.48	0.45	0.45	0.47	0.44	0.40	0.42	**0.150**
	Chinese	0.48	0.52	0.46	0.44	0.43	0.45	0.45	0.44	0.49	0.47	**0.084**
	Arabic	0.50	0.50	0.46	0.42	0.40	0.40	0.40	0.38	0.39	0.38	**0.124**
Grok 4.20	English	0.14	0.24	0.76	0.71	0.88	0.97	0.93	1.00	0.72	1.00	**0.855**
	French	0.46	0.53	0.99	0.74	0.80	0.92	0.89	1.00	0.62	1.00	**0.536**
	Vietnamese	0.26	0.74	1.00	0.72	0.99	0.99	1.00	1.00	0.72	0.98	**0.740**
	Chinese	0.39	0.43	0.69	0.43	0.54	0.77	0.77	0.69	0.65	0.66	**0.381**
	Arabic	0.12	0.26	0.62	0.55	0.84	0.90	0.70	0.70	0.55	0.69	**0.786**
Gemini 3.1 Flash-Lite	English	0.74	0.62	0.67	0.62	0.63	0.62	0.59	0.62	0.57	0.62	**0.167**
	French	0.57	0.66	0.70	0.63	0.65	0.60	0.60	0.61	0.60	0.56	**0.145**
	Vietnamese	0.58	0.56	0.54	0.64	0.57	0.55	0.57	0.49	0.47	0.47	**0.170**
	Chinese	0.62	0.66	0.53	0.61	0.59	0.60	0.60	0.58	0.58	0.58	**0.133**
	Arabic	0.69	0.74	0.66	0.70	0.74	0.69	0.71	0.69	0.69	0.66	**0.080**

Table S4: Variance decomposition of the cell means, five models. Type-II analysis of variance on the 250 model × language × scale cell means of the five models the main text reports, where the design is exactly balanced. The payoff scale carries a significant main effect, and its interaction with the model carries three times as much of the variance.

Term	Sum sq.	df	F	p	% variance
Model	4.7871	4	187.27	< 0.001	45.4
Language	0.1599	4	6.25	< 0.001	1.5
Payoff scale	0.6119	9	10.64	< 0.001	5.8
Model × language	1.7901	16	17.51	< 0.001	17.0
Model × scale	1.8910	36	8.22	< 0.001	17.9
Language × scale	0.3844	36	1.67	0.018	3.6
Residual	0.9202	144			8.7

Table S5: Variance decomposition of the cell means, all six models. The same type-II analysis on all 300 cell means, the sixth model included. Every term significant in the five-model table is significant here and in the same order: the model × scale interaction again carries several times the variance of the payoff scale as a main effect.

Term	Sum sq.	df	F	p	% variance
Model	4.8330	5	170.26	< 0.001	44.8
Language	0.1285	4	5.66	< 0.001	1.2
Payoff scale	0.4661	9	9.12	< 0.001	4.3
Model × language	1.9467	20	17.15	< 0.001	18.0
Model × scale	2.0646	45	8.08	< 0.001	19.1
Language × scale	0.3331	36	1.63	0.020	3.1
Residual	1.0219	180			9.5

Table S6: Persona effect by model and payoff scale. The cooperation rate of an agent instructed that it is cooperative minus that of an agent instructed that it is selfish, at each scale. The two leftmost columns are the scales at which every payoff printed is at most 1.

Model	λ=0.01	λ=0.1	λ=0.25	λ=0.5	λ=1	λ=2	λ=5	λ=10	λ=100	λ=1000
Claude Haiku 4.5	−0.211	−0.232	+0.206	+0.200	+0.179	+0.261	+0.277	+0.158	+0.168	+0.211
GPT-5.4 Nano	−0.172	−0.098	+0.015	−0.026	−0.053	+0.032	+0.041	+0.055	−0.017	+0.100
Gemini 3.5 Flash-Lite	−0.335	−0.578	+0.122	+0.234	+0.241	+0.258	+0.313	+0.203	+0.193	+0.230
Qwen3 235B-A22B	−0.901	−0.922	−0.811	−0.768	−0.702	−0.635	−0.690	−0.602	−0.657	−0.566
Grok 4.20	−0.454	−0.428	−0.263	−0.484	−0.323	−0.167	−0.235	−0.206	−0.515	−0.234
Gemini 3.1 Flash-Lite	+0.552	+0.602	+0.640	+0.670	+0.693	+0.681	+0.707	+0.696	+0.659	+0.692

Table S7: Share of agent-games matched exactly by one canonical rule (%). Computed by exact rule matching, with no learned component. 400 agent-games per entry.

Model	λ=0.01	λ=0.1	λ=0.25	λ=0.5	λ=1	λ=2	λ=5	λ=10	λ=100	λ=1000
Claude Haiku 4.5	20.0	15.8	3.2	3.5	4.0	2.8	4.0	2.5	4.5	3.5
GPT-5.4 Nano	12.5	7.0	9.2	6.2	5.5	6.5	8.2	7.2	8.5	9.0
Gemini 3.5 Flash-Lite	30.8	33.8	20.2	13.0	20.5	14.2	14.8	11.5	14.0	16.2
Qwen3 235B-A22B	77.0	67.2	63.0	62.7	59.8	45.2	58.0	43.2	55.0	42.0
Grok 4.20	52.5	36.8	15.0	21.2	17.5	8.8	14.8	13.8	30.0	15.5
Gemini 3.1 Flash-Lite	16.5	19.0	21.2	23.5	29.0	23.0	30.0	29.5	27.5	26.8

Table S8: Strategy composition by model and payoff scale (%). The label assigned by the hybrid read-out: the rule base fixes the candidate set and the LSTM ranks within it, deciding alone only where no canonical rule fits. Columns sum to 100 within each model and scale.

Model	Label	0.01	0.1	0.25	0.5	1	2	5	10	100	1000
Claude Haiku 4.5	AllC	27.0	20.0	14.2	20.0	21.8	18.2	15.2	18.8	12.8	14.0
	TFT	20.5	23.5	23.5	18.0	16.8	16.5	19.5	17.8	20.8	22.2
	WSLS	8.8	13.5	15.2	13.5	20.2	21.2	20.5	18.2	21.8	19.0
	AllD	43.8	43.0	47.0	48.5	41.2	44.0	44.8	45.2	44.8	44.8
GPT-5.4 Nano	AllC	29.8	41.8	52.2	46.2	48.5	54.2	58.5	52.2	53.2	50.7
	TFT	11.8	11.5	10.5	11.8	10.2	13.2	13.8	13.2	13.0	17.8
	WSLS	12.5	12.5	14.5	15.5	13.8	10.5	12.5	10.2	15.5	14.0
	AllD	46.0	34.2	22.8	26.5	27.5	22.0	15.2	24.2	18.2	17.5
Gemini 3.5 Flash-Lite	AllC	73.8	59.8	74.0	64.0	58.0	60.0	51.7	59.8	56.2	51.5
	TFT	6.8	2.2	10.2	16.2	17.0	16.2	15.2	13.5	15.2	14.2
	WSLS	6.2	13.2	14.0	16.2	21.2	12.0	18.0	18.5	16.5	15.2
	AllD	13.2	24.8	1.8	3.5	3.8	11.8	15.0	8.2	12.0	19.0
Qwen3 235B-A22B	AllC	42.0	47.8	39.2	38.0	36.8	36.5	35.5	35.2	34.5	33.8
	TFT	0.5	0.0	0.0	0.0	1.2	0.8	2.0	6.2	0.0	4.8
	WSLS	9.0	2.5	8.8	13.5	7.2	12.0	8.5	8.2	9.2	10.0
	AllD	48.5	49.8	52.0	48.5	54.8	50.7	54.0	50.2	56.2	51.5
Grok 4.20	AllC	18.0	29.8	72.8	48.8	71.0	81.5	74.8	82.0	49.5	79.0
	TFT	5.8	10.0	9.5	9.0	4.2	5.8	6.8	3.2	8.2	4.2
	WSLS	9.8	14.8	8.5	17.5	13.5	9.0	11.5	7.0	20.0	8.0
	AllD	66.5	45.5	9.2	24.8	11.2	3.8	7.0	7.8	22.2	8.8
Gemini 3.1 Flash-Lite	AllC	55.8	57.0	58.2	61.0	59.8	55.0	58.5	53.8	52.2	52.2
	TFT	6.5	3.8	2.2	0.2	2.5	3.5	1.2	2.2	6.2	5.8
	WSLS	12.8	13.2	8.8	2.8	8.0	7.2	6.5	7.2	7.2	6.0
	AllD	25.0	26.0	30.8	36.0	29.8	34.2	33.8	36.8	34.2	36.0

Table S9: Provenance of each label. For every label, the share of the agent-games carrying it that were deduced (exactly one canonical rule fits), ambiguous (several fit, and the LSTM ranked within that set) or unmatched (no canonical rule fits, so the label is an attribution the LSTM makes alone). The asymmetry is severe and bounds how the compositional result may be read: AllC and AllD carry a substantial deduced share, whereas the TFT and WSLS labels are almost entirely attributions.

Label	n	Deduced (%)	Ambiguous (%)	Unmatched (%)
AllC	11,312	23.6	36.9	39.6
TFT	2,277	5.4	0.0	94.6
WSLS	2,972	2.3	0.6	97.1
AllD	7,439	35.6	0.0	64.4

Table S10: Validation of the LSTM branch. Trained on synthetic AllC / AllD / Tit-for-Tat / Win-Stay-Lose-Shift trajectories at zero and 5% execution noise, balanced at 40,320 sequences per strategy per level. The 10% and 20% noise levels are never trained on. No language-model transcript enters training.

Split	n	Accuracy
held-out (NoNoise + Noise005)	64,512	0.9846
OOD (Noise01)	161,280	0.9364
OOD (Noise02)	161,280	0.8107
held-out recall: AllC	16,150	0.9853
held-out recall: AllD	16,203	0.9952
held-out recall: TFT	16,193	0.9744
held-out recall: WSLS	15,966	0.9834

Table S11: Pairwise correlation between the model curves, five models. Correlation of the ten-point cooperation-against-scale curves of each pair of the five models the main text reports. Six of the ten pairs are negative and the mean is −0.10, which is why averaging the curves cancels much of the movement.

	Claude Haiku	GPT-5.4 Nano	Gemini 3.5	Qwen3 235B-A22B	Grok 4.20
Claude Haiku 4.5		+0.11	−0.18	−0.56	+0.40
GPT-5.4 Nano	+0.11		−0.23	−0.72	+0.85
Gemini 3.5 Flash-Lite	−0.18	−0.23		+0.22	−0.15
Qwen3 235B-A22B	−0.56	−0.72	+0.22		−0.71
Grok 4.20	+0.40	+0.85	−0.15	−0.71	

Table S12: Pairwise correlation between the model curves, all six models. The same correlations with the sixth model included: nine of the fifteen pairs are negative and the mean is −0.08.

	Claude Haiku	GPT-5.4 Nano	Gemini 3.5	Qwen3 235B-A22B	Grok 4.20	Gemini 3.1
Claude Haiku 4.5		+0.11	−0.18	−0.56	+0.40	−0.03
GPT-5.4 Nano	+0.11		−0.23	−0.72	+0.85	−0.70
Gemini 3.5 Flash-Lite	−0.18	−0.23		+0.22	−0.15	+0.36
Qwen3 235B-A22B	−0.56	−0.72	+0.22		−0.71	+0.73
Grok 4.20	+0.40	+0.85	−0.15	−0.71		−0.57
Gemini 3.1 Flash-Lite	−0.03	−0.70	+0.36	+0.73	−0.57	